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ISO 9001* AQAP 2110*
ISO/IEC 22301
*GDPR *ISO 14001

Design Document

for the

EasyShare System

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Content

1. System Overview

EasyShare is a web-based file management system that allows users to efficiently manage digital content through folders and files. It supports user authentication, file organization, access control, and administrative file recovery in a user-friendly interface.

2. System Architecture

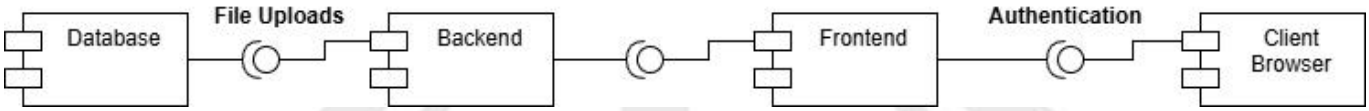

Diagram 1: High-Level Component Diagram

- **Frontend**: HTML, CSS (Bootstrap 5), JavaScript (vanilla or framework-based)
- **Backend**: Node.js with Express.js
- **Database**: PostgreSQL or MongoDB (for flexibility in file structure)
- **Authentication**: Session-based or JWT
- **File Uploads**: Multer for handling file uploads

3. Functional Module Designs

3.1. Login and Authentication

- Login Page: Username and password input
- Backend Logic:
  - Validate credentials against the database
  - Establish session or return JWT token
  - Role detection (e.g., admin vs standard user)

- Security:
  - Passwords hashed using bcryptjs
  - CSRF and XSS protections enabled

3.2. Home Page Functions

- Folder Management
  - Create Folder: User inputs a name; backend stores metadata and creates logical path
  - Remove Folder: Marks folder as deleted (soft delete)
  - Search Folder: Filters folders by name using search input
  - Set Visibility:
    - public, private (role-based access control)
    - Stored as part of folder metadata
- File and Subfolder Management
  - Upload File / Add Subfolder: Files uploaded using Multer and linked to a parent folder ID
  - Move File / Subfolder:
    - Drag-and-drop UI or dropdown selector
    - Backend updates the parent folder reference
  - Remove File: Soft delete with timestamp
  - Search Files/Subfolders:

- Real-time search using input field
- Case-insensitive partial matching

❖ Set Visibility Rights:

- Each file or subfolder can have its own access control
- Admins can override access restrictions

➢ Deleted Files Management (Admin Only)

❖ Recycle Cash View:

- Displays deleted files and folders
- Only accessible by system administrators

❖ Restore File:

- Reverts the deleted flag and returns file to location chose by user

3.3. Help Page

➢ Static or dynamically loaded help content

➢ Sections:

❖ Main Page

❖ Helper

4. Database Design (Simplified ERD)

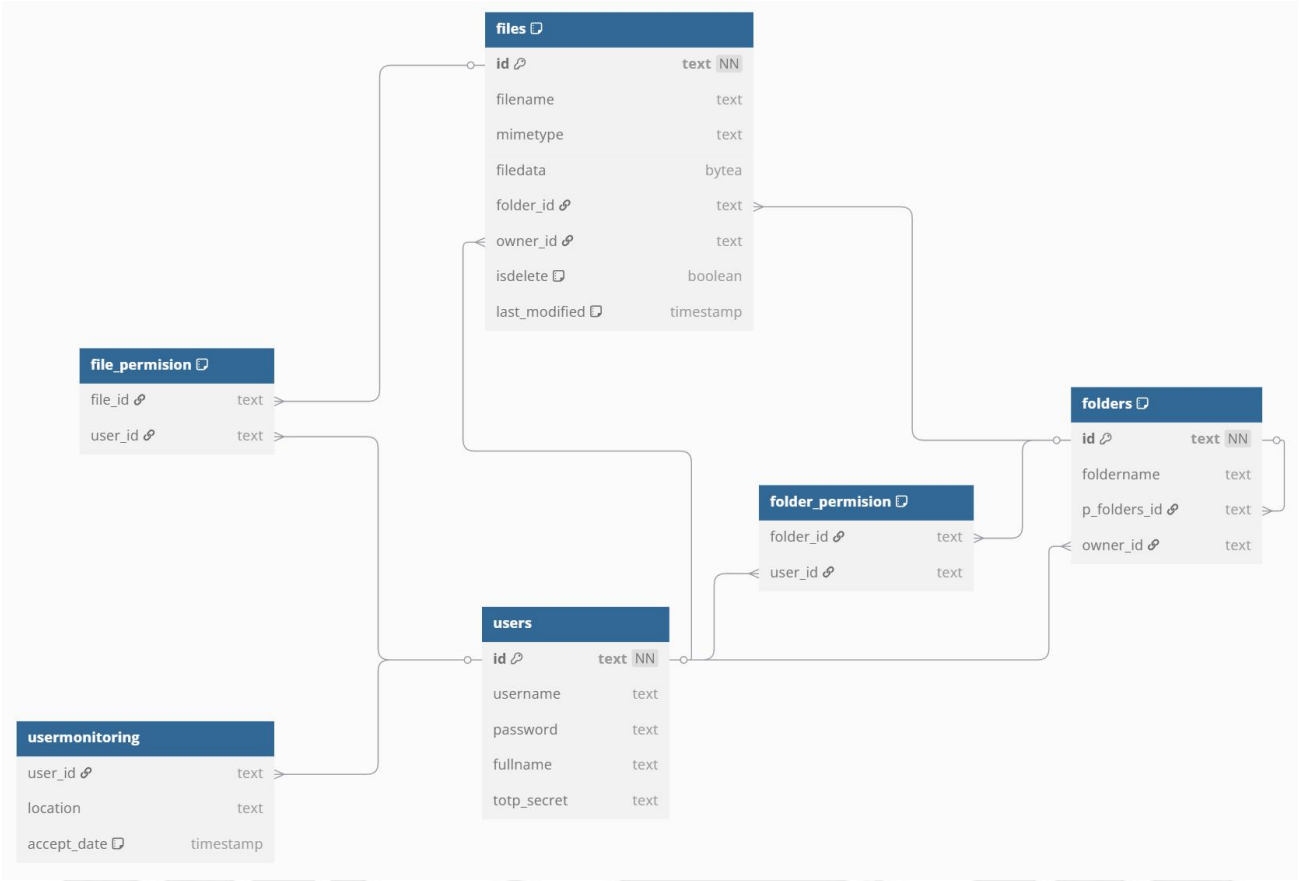

Diagram 2: Entity-Relationship Diagram

5. User Interface Design

5.1. Navigation Layout:

- Top navbar: Main page | Helper | Log out
- Main page:
  - List of all root folders
  - Create/Delete Folder button
  - Cash button
  - Search and Filter tools

5.2. Modal Dialogs:

- Current folder content
- Create/Move/Delete Sub-Folders
- Upload/Move/Delete Files
- Set Visibility
- Confirm Delete or Restore

6. Permissions Model

Action	User	Admin
View folders/files	✔	✔
Add folders/files	✔	✔
Delete folders/files	✔	✔
Move folders/files	✔	✔
Restore deleted files	✘	✔
Set visibility rights	✔	✔
Access Help	✔	✔

Table 1: Permissions model

7. Design Considerations

- **Scalability**: Folder hierarchy can be nested deeply; recursive querying supported.
- **Performance**: Indexed searches; lazy loading for folder contents.
- **Accessibility**: ARIA labels and keyboard support for major actions.
- **Responsiveness**: Layout adapts to mobile and tablet screen sizes.